

Course: B. Tech. Branch: Information Technology Semester: VII

Subject Code & Name: BTITPE705E Data Warehousing and Data Mining

Max Marks: 60

Date:09/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- | | | |
|---|------|---|
| A) What is Data Warehousing? Explain features of data warehouse. | CO 1 | 6 |
| B) Illustrate the architecture of a typical Data warehouse System. | CO 1 | 6 |
| C) Suppose that a data warehouse consists of the three dimensions time, doctor and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.
(i) Draw a star schema diagram for the above data warehouse.
(ii) Starting with the base cuboid [day, doctor, patient], what specific OLAP operations should be performed in order to list the total fee collected by each doctor in 2004?
(iii) To obtain the same list, write an SQL query assuming the data are stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge). | CO 2 | 6 |

Q.2 Solve Any Two of the following.

12

- | | | |
|---|------|---|
| A) Describe categories of OLAP Tools. | CO 2 | 6 |
| B) Explain Data mining as a step in KDD. Give the architecture of typical data mining system. | CO 1 | 6 |
| C) Suppose that the data for analysis includes the attribute salary. We have the following values for salary (in thousands of dollars), shown in increasing order: 30, 36,47,50,52,52,56,60,63,70,70,110.
(i) What are the mean, median, mode and midrange of the data?
(ii) Find the first quartile (Q1) and the third quartile (Q3) of the data.
(iii) Show a boxplot of the data. | CO 2 | 6 |

Q. 3 Solve Any Two of the following. **12**

- A) Discuss different steps involved in Data Preprocessing. CO 2 **6**
- B) What is clustering? Explain K-means clustering algorithm. CO 3 **6**
- C) Suppose the data for clustering is {2, 4, 10, 12, 3, 20, 30, 11, 25, 5, 36, 41, 14}. CO 3 **6**
Assuming number of clusters to be 2 i.e. $K = 2$, cluster the given data using above algorithm.

Q.4 Solve Any Two of the following. **12**

- A) Differentiate between Classification, Prediction and Clustering. CO 3 **6**
- B) Given is the training data for height classification, classify the tuple $t = \langle \text{Arvish}, M, 1.97 \rangle$ using Bayesian classification. CO 3 **6**

Name	Gender	Height	Output
Reena	F	1.6 m	Short
Mahesh	M	2 m	Tall
Tina	F	1.9 m	Medium
Meeta	F	1.88 m	Medium
Siya	F	1.7 m	Short
Vikram	M	1.85 m	Medium
Lakshmi	F	1.6 m	Short
Andrew	M	1.7 m	Short
Henry	M	2.2 m	Tall
Akhil	M	2.1 m	Tall
Lata	F	1.8 m	Medium
Siraj	M	1.95 m	Medium
Rita	F	1.9 m	Medium
Kriti	F	1.8 m	Medium
Srishti	F	1.75 m	Medium

- C) Explain single linkage, average linkage and complete linkage algorithms. CO 3 **6**

Q. 5 Solve Any One of the following. **12**

- A) Discuss Association Rule Mining and Apriori Algorithm. CO 3 **12**
Consider the transaction database given below,

TID	Items
10	1,3,4
20	2,3,5
30	1,2,3,5
40	2,5
50	1,3,5

Use Apriori Algorithm with min-support count =2 and min-confidence =60% to find all frequent itemsets and strong association rules.

- B) a)** What is Web Structure Mining? Explain Page Rank technique in detail.
b) Write about case study on Mining Text Databases.

CO 4

12

*** End ***